CS & IT ENGINEERING



Algorithms

Greedy Method

Lecture No.- 04



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Job Sequencing with Deadline

Optimal Merge Patterns (OMP)

Topics to be Covered



- Topic
- Topic — Optimal Merge Pattern
- Topic — Huffman Encoding

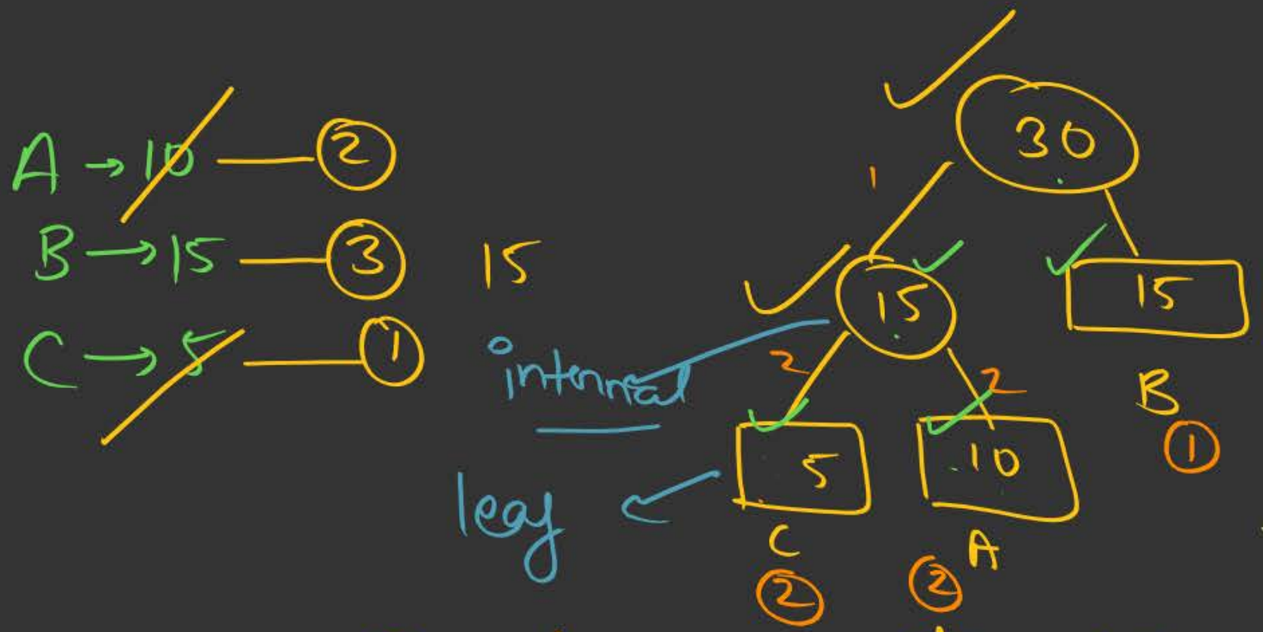


About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW



Total Record movements = 15 + 30

= 45

Sum of internal nodes of 2-way Optimal Binary merge Tree (OBMT)



Topic : Greedy of Algorithms

Optimal merge pattern

Eg.1. A : 10.....(2)

B : 3 .....(3)

C : 5.....(1)

2. Formal => (Weighted external path length) of the OBMT
Mino record movement Σdi *qi
= 2×10 + 1×15 + 2 × 5 = 20 + 15 + 10 = 45

$\Sigma d_i q_i$

$d_i \rightarrow$ depth $\rightarrow$ Dist of File f_i from root $\rightarrow$ no. of edges

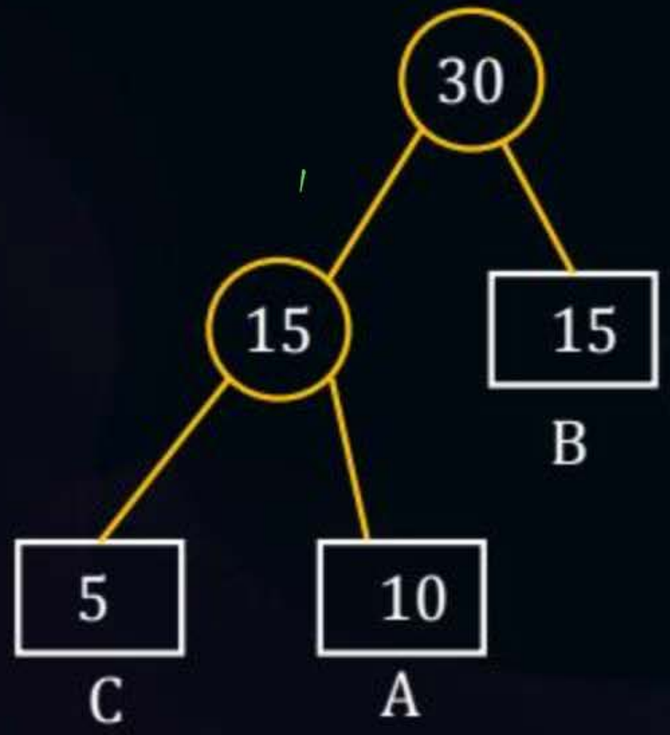
$q_i \rightarrow$ File Size

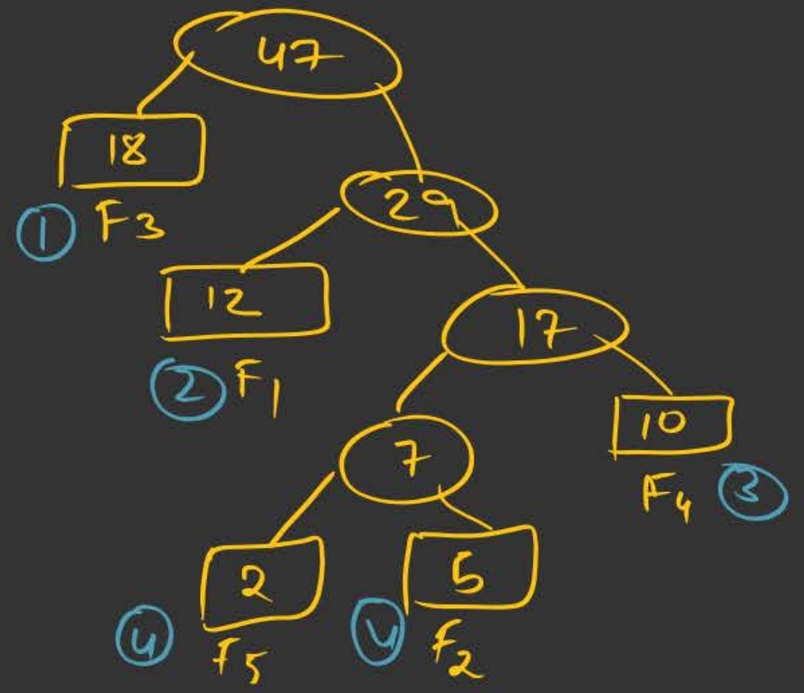


Topic : Greedy Algorithms

Optimal 2-way Binary merge Tree

Min total record movement = 15 + 30 = 45





85%
69%

F1 – 12 — 4
F2 – 5 — 2
29, F3 = 18 — 5
F4 = 10 — 3
F5 = 7 — 1

min RM =

RM = 47 + 29 + 17 + 7

= 100

RM = $\sum d_i q_i$

= 18×1 + 12×2 + 10×3 + 2×4 + 5×4

= 18 + 24 + 30 + 8 + 20

= 100



Topic : Greedy Algorithms

Greedy algo for optimal merge patterns

```
tree node = record
{
        tree node * lchild;
        tree nodes * rchild ;
        int weight;
}
```



Topic : Greedy Algorithms

```
Algo tree (n)
{
    // list is a global list of n single nodes binary tree, as described before
    for i : 1 to (n-1) ;→ O(n)
    {
        1. Pt = new (treenode)
        2. pt → lchild = least (list)
        3. pt→ rchild = least (list)
        4. pt → weight = (pt→ lchild) → weight
                         ((pt → rchild) → weight
        5. insert (list, pt)
    }
```





Topic : Greedy Algorithms

Time complexity: (for 'n' files)

1. Linear linked list:

('n' elements)

Insert → O(1)

least → O(n)

Overall TC; $O(n^2)$

min

$n! \rightarrow n^2$



Topic : Greedy Algorithms

Implementation of list

2. Heap (non-linear)

('n' elements)

Insert → logn

Delete → logn

~~Dec./Inc.~~

Min Heap: least → O(1)

Overall TC: O(n*logn)

n! → nlogn

Huffman chachaji



Huffman Encoding



Topic : Greedy Algorithms

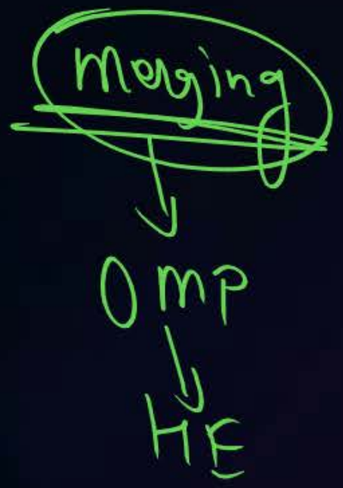
3. Huffman Encoding

(Application of optimal merge pattern)

(Data encoding and data compression Technique)

(representing some elements with a unique code)

Compression→ (reduce size of data without data loss)



① Binary Encoding





Topic : Greedy Algorithms

Eg.1. 4 Character /elements

w, x, y, z

Straight forward Binary encoding

Normal Encoding

Uniform Encoding

W → 0 0

X → 0 1

Y → 1 0

Z → 1 1

10 dem



Topic : Greedy Algorithms

5 → 3
6 → 3
8 → 3

If we use uniform encoding:

1. 6 elements → 3 bits
2. 10 elements → 4 bit → $\log_2 10 \rightarrow \lceil 3.x \rceil$
   → 4

   4 bits → max $2^4 = 16$ elements can be uniquely encoded
3. 17 element → 5 bits.....
4. 100 elements → 7 bits / elements or character.

(5 - 8)
(9 - 16) —

$\log_2 17 = 5$



Topic : Greedy Algorithms

In general

To represent N elements using uniform encoding

we need → $\lceil \log_2 (N) \rceil$ bit/character.

Imp. Observations:

1. Using uniform encoding, data compression is not possible.
2. Hence, for data compression we need non-uniform encoding.



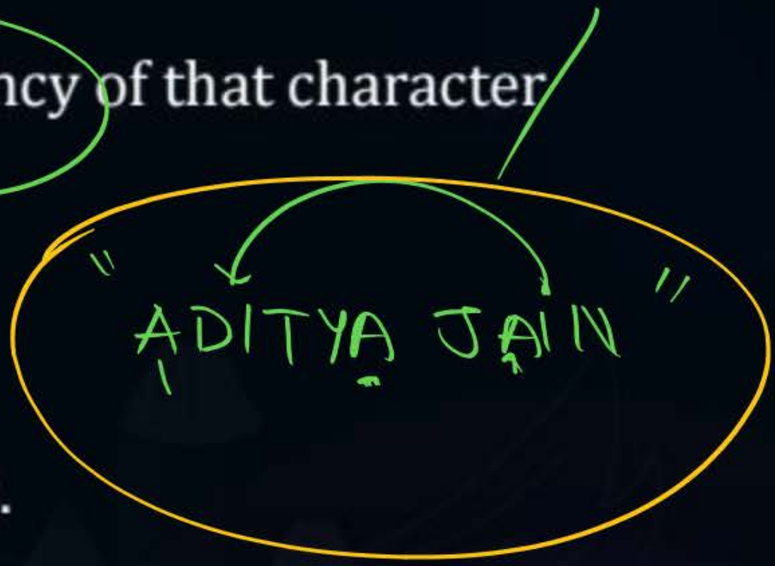
Topic : Greedy Algorithms

Huffman Encoding (non-uniform encoding)

Idea:- Encoding/code → assigned based on the frequency of that character elements.

Logic:- $$\text{(no. of bits to encode)} \propto \frac{1}{\text{(freq.of character)}}$$

1. Assign more bits to low frequently occurring elements.
2. Assign less bits to high frequently occurring elements.
3. No. 2 elements/characters will have same encoding (as it needs to be unique).





Topic : Greedy Algorithms

Eg..

Given: Language/ Elements $\Rightarrow$ {A,B,C,D,E} $\rightarrow$ 5 character

If uniform encoding: $\lceil \log_2 N \rceil$ bits character

$= \lceil \log_2 5 \rceil$

= 3 bits/ character

3 > 1.95

Note:- Start from lowest level.



Topic : Greedy Algorithms

eg

A → 0.23.....(4)

B → 0.05.....(1)

C → 0.48.....(5)

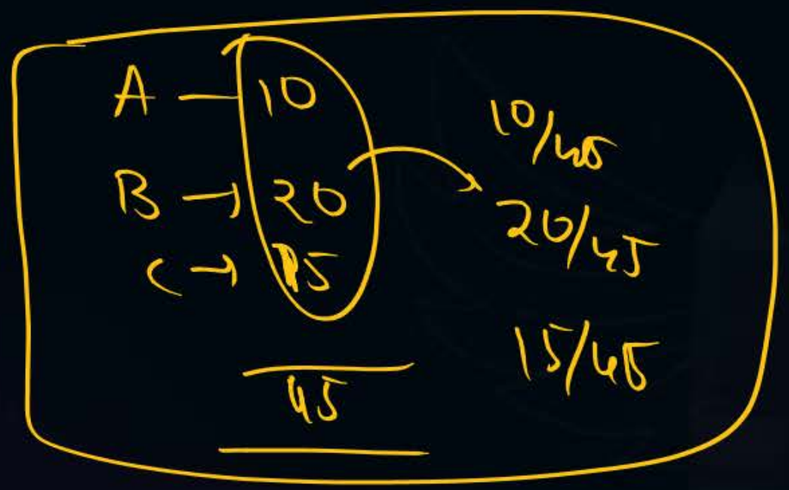
D → 0.15.....(3)

E → 0.09.....(2)

(Probability / frequency)



Priority (low to high)





Topic : Greedy Algorithms

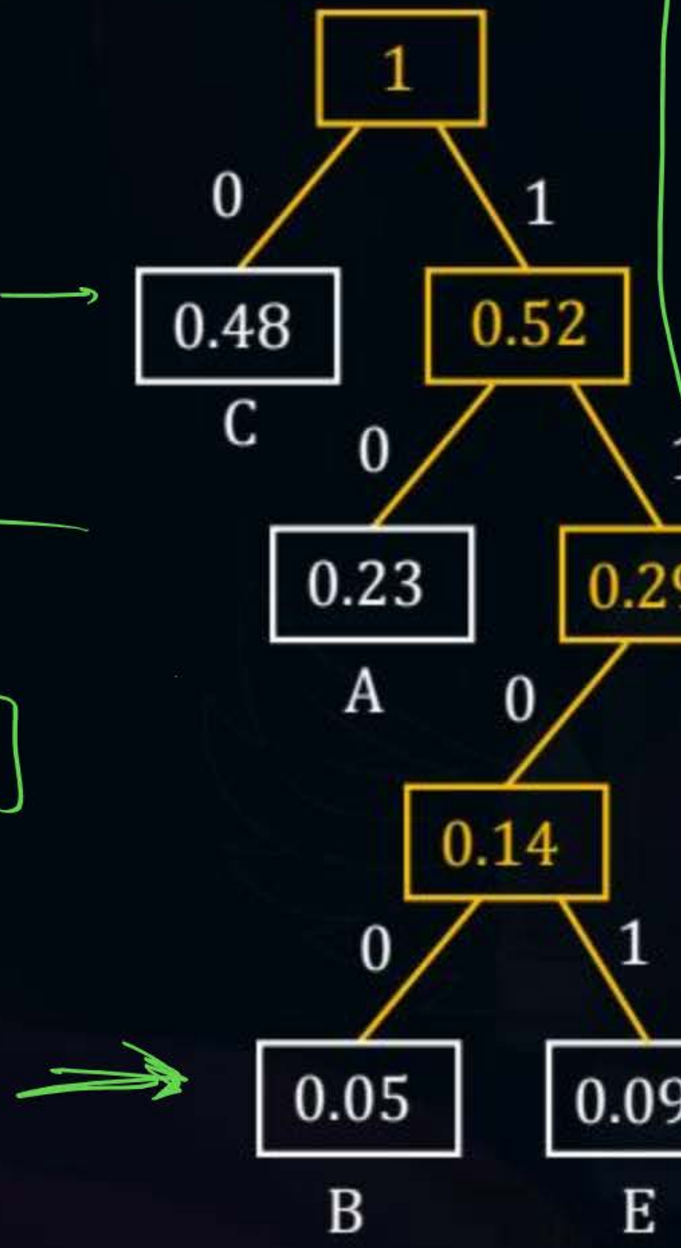
2-way Optimal Binary merge tree:

always
- Left Branch → 0
- Right Branch → 1

C → 0
A → 10

Encoding
[Root → Node]

Standard
- Left → Smaller
- Right → Larger



1+0.52
+0.29
+0.14
1.52
+0.43
1.95



Topic : Greedy Algorithms

A → 10 → 2 bits

B → 1100 → 4 bits

C → 0 → 1 bits

D → 111 → 3 bits

E → 1101 → 4 bits

Non- Uniform encoding & all are unique



Topic : Greedy Algorithms

#Q.1) ~~Averae~~ number of bit required using Huffman encoding? Coding:

Average

= Weighted external path length:

$= \sum_{i=1}^{n} di * qi$

depth → freq of char(i)

= Ans :- 1.95 bits/char



Topic : Greedy Algorithms

text =

#Q.2) "C C A C C C A B D C"

(Use data of prev ques)

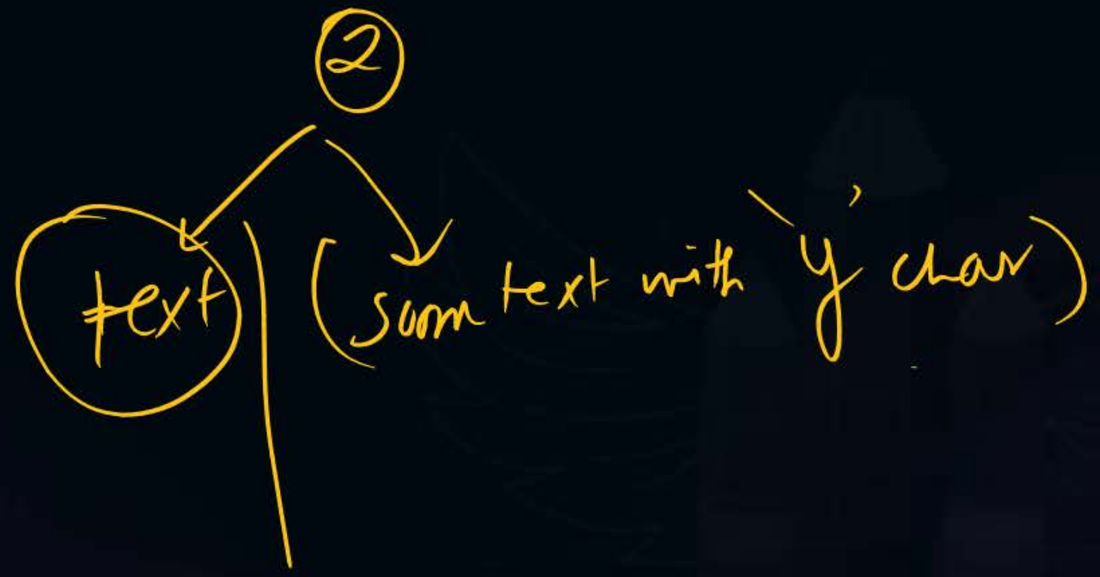
A → 2 → 2×3
B → 1 → 1×3
C → 6 → 6×3
D → 1 → 1×3
E → 0 → 0×3

18.5 %

How many bit required to above text using:- encode

1. Uniform Encoding:
2. Huffman Encoding:

(Non-uniform)



① Total bits using uniform encoding

= 3×10 = 30 bits



Topic : Greedy Algorithms

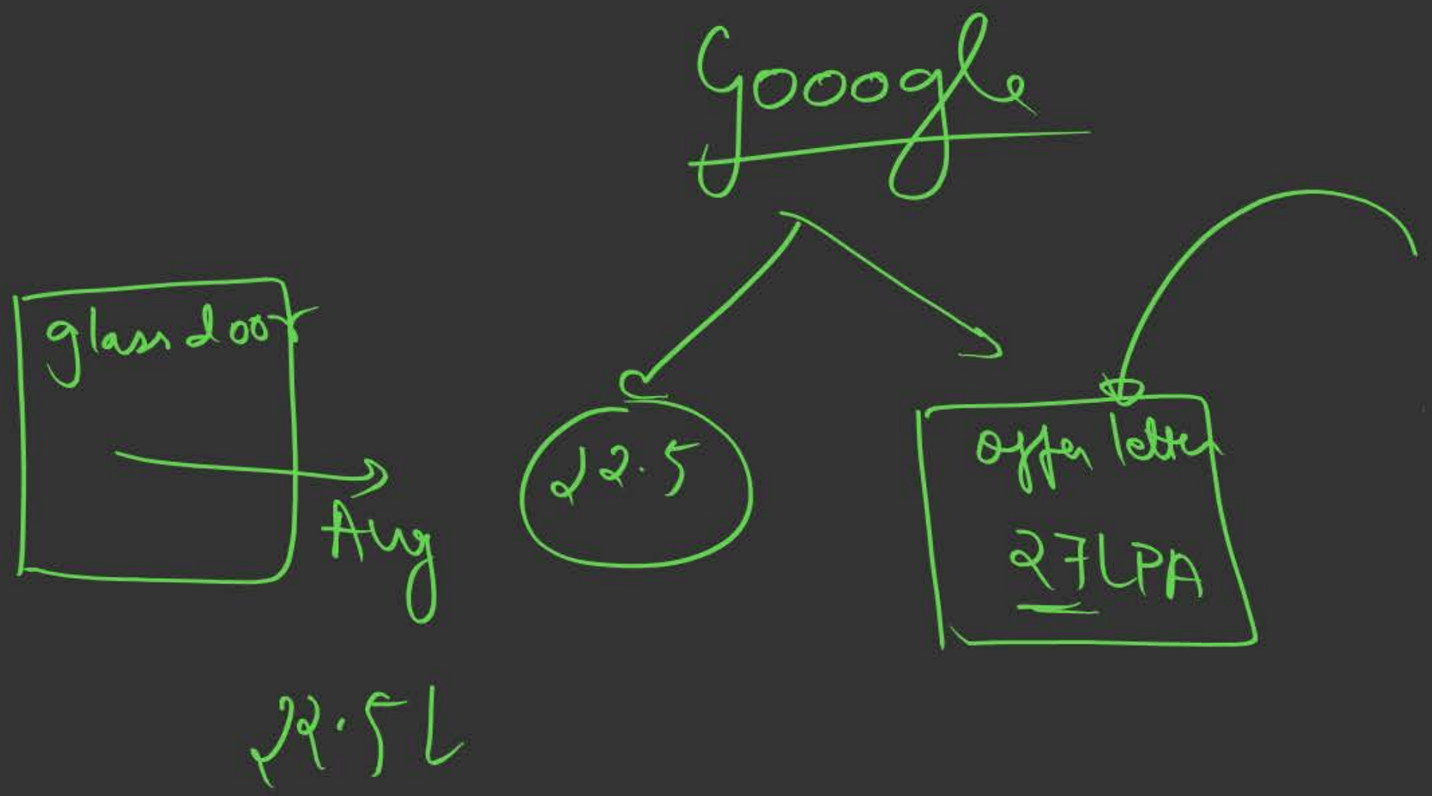
2. Using Huffman Encoding:

Text:	C	C	A	C	C	C	A	B	D	C
	0	0	10	0	0	0	10	1100	111	0

Final Encoding:

0 0 1 0 0 0 0 1 0 1 1 0 0 1 1 1 0

Total = 17 bit compressions


Gooogle
glassdoor
Avg
22.5
offer letter
27LPA
22.5L

Extension of Prev

Type (9.3)

Some txt of 100 char is to be encoded using most optimal encoding technique.

Then what are the total no. of bits reqd to encode such a message?

(Huffman Encoding)

32%

Floor

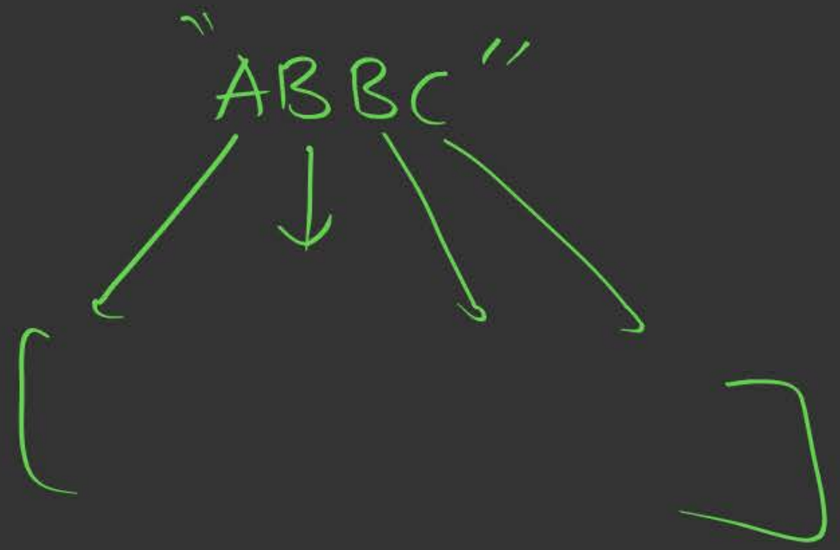
$$\text{Ans} = (\text{Avg. no of bits/char}) * \text{no. of char}$$

$$= 1.95 \times 100 = 195 \text{ bits}$$

Extension

Type-4 :- Encode a message

H.W





Topic : Greedy Algorithms

H.W

#Q. The characters 'A' to 'H' have the set of frequencies based on the first 8 Fibonacci number as follows:

A =1, B =1, C=2, D=3, E= 5, F= 8, G = 13, H=21

A Huffman code is used to represent the characters. What is the sequence of characters corresponding to the following code: 11011110011010

A FDHEG

B ECGDF

C DCHFG

D FEHDG



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

OMP
HE



THANK - YOU